EE4-66

Solution of Question 1.

(a) The objective function f_0 is convex. h_i's are convex. ℓ_i's are affine. [3]

(b) The Lagrange dual function is given by $L(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{v}) = f(\boldsymbol{x}) + \sum_{i=1}^{m} u_i h_i(\boldsymbol{x}) + \sum_{j=1}^{r} v_j \ell_j(\boldsymbol{x})$. where $\boldsymbol{u} \in \mathbb{R}^m$, $\boldsymbol{v} \in \mathbb{R}^r$, and $\boldsymbol{u} \geq \boldsymbol{0}$. [2]

(c) The KKT conditions are given by

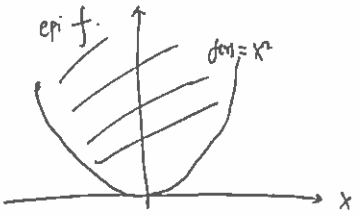
- $0 \in \partial f(\boldsymbol{x}) + \sum_{i=1}^{m} u_i \partial h_i(\boldsymbol{x}) + \sum_{j=1}^{r} v_j \partial \ell_j(\boldsymbol{x})$.
- $u_i h_i(\boldsymbol{x}) = 0$, $\forall i$.
- $h_i(\boldsymbol{x}) \leq 0$, $\ell_j(\boldsymbol{x}) = 0$, $\forall i$, $\forall j$.
- $u_i \geq 0$, $\forall i$. [2]

(d) By KKT conditions, $\boldsymbol{x}$ is the minimizer if and only if

$$\begin{bmatrix} \boldsymbol{Q} & \boldsymbol{A}^T \\ \boldsymbol{A} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{v} \end{bmatrix} = \begin{bmatrix} -\boldsymbol{c} \\ \boldsymbol{0} \end{bmatrix},$$

where the first set of linear equations come from the stationarity and the second set follows from the primal feasibility. The optimal $\boldsymbol{x}^\star$ can be obtained by solving the linear inverse problem. [3]

(e)



[2]

(f) Let $(\boldsymbol{u}, a), (\boldsymbol{v}, b) \in \text{epi} f$. Then $\forall \lambda \in [0,1]$ by the convexity of f,

$$f(\lambda \boldsymbol{u} + (1-\lambda)\boldsymbol{v}) \leq \lambda f(\boldsymbol{u}) + (1-\lambda) f(\boldsymbol{v}) \leq \lambda a + (1-\lambda) b.$$

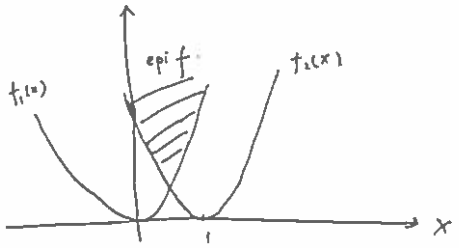
Hence $\lambda(\boldsymbol{u}, a) + (1-\lambda)(\boldsymbol{v}, b) \in \text{epi} f$. [3]

(g)

$$\begin{aligned}
& f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \\
&= \max\{f_1(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}),\ f_2(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y})\} \\
&\leq \max\{\lambda f_1(\boldsymbol{x}) + (1-\lambda) f_1(\boldsymbol{y}),\ \lambda f_2(\boldsymbol{x}) + (1-\lambda) f_2(\boldsymbol{y})\} \\
&\leq \lambda \max\{f_1(\boldsymbol{x}), f_2(\boldsymbol{x})\} + (1-\lambda)\max\{f_1(\boldsymbol{y}), f_2(\boldsymbol{y})\} \\
&= \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).
\end{aligned}$$

[3]

(h)



[2]

Solution of Question 2.

(a) The spark of $\boldsymbol{A}$ is the minimum number of linearly dependent columns in $\boldsymbol{A}$. [2]

i $\operatorname{spark}(\boldsymbol{A}) = m + 1$ most likely. [1]

(b)

$$\mu = \max_{i \neq j} |\langle \boldsymbol{a}_i, \boldsymbol{a}_j \rangle|,$$

where $\boldsymbol{a}_i$ and $\boldsymbol{a}_j$ are the i-th and j-th columns of $\boldsymbol{A}$ respectively. [2]

(c) The matrix $\boldsymbol{A} \in \mathbb{R}^{m \times N}$ satisfies the RIP with parameters (S, δ) for $S \leq m$ and $\delta \in (0, 1)$, if for all index sets $\mathcal{I} \subset \{1, \cdots, N\}$ such that $|\mathcal{I}| \leq S$ and for all $\boldsymbol{q} \in \mathbb{R}^{|\mathcal{I}|}$, it holds that

$$(1 - \delta) \|\boldsymbol{q}\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{I}} \boldsymbol{q}\|_2^2 \leq (1 + \delta) \|\boldsymbol{q}\|_2^2.$$

The RIC δ_S is the infimum of all parameters δ for which the RIP holds, i.e.,

$$\delta_S = \inf \left\{ \delta : \ (1 - \delta) \|\boldsymbol{q}\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{I}} \boldsymbol{q}\|_2^2 \leq (1 + \delta) \|\boldsymbol{q}\|_2^2. \right.$$
$$\left. \forall |\mathcal{I}| \leq S, \ \forall \boldsymbol{q} \in \mathbb{R}^{|\mathcal{I}|} \right\}.$$

[4]

(d)

i $\lambda(\boldsymbol{A}_{\mathcal{T}}) \in \left[\sqrt{1 - \delta}, \sqrt{1 + \delta}\right]$. [2]

ii

$$\hat{\boldsymbol{x}}_{\mathcal{T}_0} = \left(\boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{A}_{\mathcal{T}_0}\right)^{-1} \boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{y}, \ \hat{\boldsymbol{x}}_{\mathcal{T}_0^c} = \boldsymbol{0}.$$

[2]

iii It is clear that

$$\begin{aligned} &\|\boldsymbol{x}_0 - \hat{\boldsymbol{x}}\|_2 \\ &= \left\| \boldsymbol{x}_{0,\mathcal{T}_0} - \left(\boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{A}_{\mathcal{T}_0}\right)^{-1} \boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{A}_{\mathcal{T}_0} \hat{\boldsymbol{x}}_{\mathcal{T}_0} - \left(\boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{A}_{\mathcal{T}_0}\right)^{-1} \boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{e} \right\|_2 \\ &= \left\| \left(\boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{A}_{\mathcal{T}_0}\right)^{-1} \boldsymbol{A}_{\mathcal{T}_0}^T \boldsymbol{e} \right\|_2. \end{aligned}$$

One has

$$\frac{\sqrt{1 - \delta_S}}{1 + \delta_S} \epsilon \leq \|\boldsymbol{x}_0 - \hat{\boldsymbol{x}}\|_2 \leq \frac{\sqrt{1 + \delta_S}}{1 - \delta_S} \epsilon.$$

Tighter bounds can also be obtained:

$$\frac{1}{\sqrt{1+\delta_S}}\epsilon \leq \|\boldsymbol{x}_0 - \hat{\boldsymbol{x}}\|_2 \leq \frac{1}{\sqrt{1-\delta_S}}\epsilon.$$

[4]

iv Note that

$$2\left(1-\delta_{2S}\right) \leq \|\boldsymbol{A}_{\mathcal{I}}\boldsymbol{a} + \boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\|_2^2 = \left\|[\boldsymbol{A}_{\mathcal{I}}\boldsymbol{A}_{\mathcal{J}}]\begin{bmatrix}\boldsymbol{a}\\ \boldsymbol{b}\end{bmatrix}\right\|_2^2 \leq 2\left(1+\delta_{2S}\right),$$

$$2\left(1-\delta_{2S}\right) \leq \|\boldsymbol{A}_{\mathcal{I}}\boldsymbol{a} - \boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\|_2^2 = \left\|[\boldsymbol{A}_{\mathcal{I}}\boldsymbol{A}_{\mathcal{J}}]\begin{bmatrix}\boldsymbol{a}\\ -\boldsymbol{b}\end{bmatrix}\right\|_2^2 \leq 2\left(1+\delta_{2S}\right).$$

As a result,

$$\langle \boldsymbol{A}_{\mathcal{I}}\boldsymbol{a}, \boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\rangle = \frac{\|\boldsymbol{A}_{\mathcal{I}}\boldsymbol{a}+\boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\|_2^2 - \|\boldsymbol{A}_{\mathcal{I}}\boldsymbol{a}-\boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\|_2^2}{4} \leq \delta_{2S},$$

$$-\langle \boldsymbol{A}_{\mathcal{I}}\boldsymbol{a}, \boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\rangle = \frac{\|\boldsymbol{A}_{\mathcal{I}}\boldsymbol{a}-\boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\|_2^2 - \|\boldsymbol{A}_{\mathcal{I}}\boldsymbol{a}+\boldsymbol{A}_{\mathcal{J}}\boldsymbol{b}\|_2^2}{4} \leq \delta_{2S}.$$

which completes the proof. [3]

Solution of Question 3.

(a)

i Let $Y' := \frac{Y}{\sigma}, X' := \frac{X}{\sigma}$, and $W' := \frac{W}{\sigma}$.

$$\begin{aligned}
&\mathrm{E}_{X,W}\left[(\eta(Y;\theta) - X)^2\right] \\
&= \sigma^2 \mathrm{E}_{X,W}\left[\left(\eta\left(\frac{Y}{\sigma};\frac{\theta}{\sigma}\right) - \frac{X}{\sigma}\right)^2\right] \\
&= \sigma^2 \mathrm{E}_{X',W'}\left[(\eta(Y';\alpha) - X')^2\right] \\
&= \sigma^2 \mathrm{mse}(1;\alpha),
\end{aligned}$$

where the last equality follows from the fact that $p_X = p_{X'}$. [3]

ii

A. When $x = \infty$, $Y = x + W - \alpha$. Hence

$$\begin{aligned}
&\mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 \,|\, x = \infty\right] \\
&= \mathrm{E}_W\left[W^2 - 2\alpha W + \alpha^2 | x = \infty\right] \\
&= 1 + \alpha^2.
\end{aligned}$$

[3]

B. When $x = 0$ and $|w| \leq \alpha$, $\eta(Y;\alpha) = 0$ and $\eta(Y;\alpha) - X = 0$. Hence $\mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 \,|\, |w| \leq \alpha, x = 0\right] = 0$. [2]

C. The key point of the sub-problem is to use the conditional probability density function for the computation:

$$\begin{aligned}
&\mathrm{E}_W[\alpha | w > \alpha] = \int_\alpha^\infty \alpha \frac{\phi(w)}{p_W(w > \alpha)} dw \\
&= \alpha \frac{p_W(w > \alpha)}{p_W(w > \alpha)} = \alpha.
\end{aligned}$$

[2]

D. It is clear that

$$\begin{aligned}
\mathrm{mse}(1;\alpha) &= \mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 | X = 0\right] p(X = 0) \\
&\quad + \mathrm{E}_W\left[(\eta(Y;\alpha) - X)^2 | X \neq 0\right] p(X \neq 0) \\
&= (1 - \epsilon) c_2 + \epsilon c_1.
\end{aligned}$$

[2]

(b)

i It is clear that

$$\begin{aligned}
&\boldsymbol{x}^t + \boldsymbol{A}^T\left(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^t\right) \\
&= \boldsymbol{x}_0 - \boldsymbol{x}_0 + \boldsymbol{x}^t + \boldsymbol{A}^T\left(\boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{w} - \boldsymbol{A}\boldsymbol{x}^t\right) \\
&= \boldsymbol{x}_0 + \left(\boldsymbol{A}^T\boldsymbol{A} - \boldsymbol{I}\right)\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right) + \boldsymbol{A}^T\boldsymbol{w}.
\end{aligned}$$

[2]

ii The i-th entry of $\boldsymbol{A}^T\boldsymbol{w}$ is given by $\boldsymbol{a}_i^T\boldsymbol{w} = \sum_{\ell=1}^m A_{\ell,i}w_\ell$. $\mathrm{E}\left[\left(\boldsymbol{a}_i^T\boldsymbol{w}\right)^2\right] = \mathrm{E}\left[\left(\sum_{\ell=1}^m A_{\ell,i}w_\ell\right)^2\right] = \mathrm{E}\left[\sum_{\ell=1}^m \left(A_{\ell,i}w_\ell\right)^2\right] = \sum_{\ell=1}^m w_\ell^2\mathrm{E}\left[A_{\ell,i}^2\right] = \frac{1}{m}\left\|\boldsymbol{w}\right\|_2^2$. Further, $\mathrm{E}\left[\left(\boldsymbol{a}_i^T\boldsymbol{w}\right)\left(\boldsymbol{a}_j^T\boldsymbol{w}\right)\right] = \boldsymbol{w}^T\mathrm{E}\left[\boldsymbol{a}_i^T\boldsymbol{a}_j\right]\boldsymbol{w} = 0$. Hence, $\boldsymbol{A}^T\boldsymbol{w} \sim \mathcal{N}\left(\boldsymbol{0}, \frac{1}{m}\left\|\boldsymbol{w}\right\|_2^2\boldsymbol{I}\right)$. [2]

iii The i-th element of $\boldsymbol{B}\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right)$ is given by $\sum_j B_{i,j}\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right)_j$. Its variance is given by $\sum_j \frac{1}{m}\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right)_j^2 = \frac{n}{m}\tau_t$. [2]

iv Note that then entries of $\left(\boldsymbol{A}^T\boldsymbol{A} - \boldsymbol{I}\right)\left(\boldsymbol{x}_0 - \boldsymbol{x}^t\right) + \boldsymbol{A}^T\boldsymbol{w}$ is approximately $\mathcal{N}\left(0, \frac{1}{\delta}\tau_t + \sigma^2\right)$.

$$\tau_{t+1} = \mathrm{mse}\left(1; \alpha\right)\left(\frac{1}{\delta}\tau_t + \sigma^2\right).$$

[2]

Solution of Question 4.

(a) The OMP algorithm is given by

Initialization:

$\boldsymbol{x} = \mathbf{0}$, $\mathcal{T}^\ell = \phi$, and $\boldsymbol{y}_r = \boldsymbol{y}$.

Iteration: $\ell = 1, 2, \cdots, S$

- $\mathcal{T}^\ell = \mathcal{T}^{\ell-1} \bigcup \text{supp}\left(H_1\left(\boldsymbol{A}^T \boldsymbol{y}_r\right)\right)$.
- $\boldsymbol{x}_{\mathcal{T}^\ell} = \boldsymbol{A}_{\mathcal{T}^\ell}^\dagger \boldsymbol{y}$.
- $\boldsymbol{y}_r = \boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}$. [3]

i For $i \notin \mathcal{T}_0$,

$$\begin{aligned} |\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| &= \left|\sum_{j\in\mathcal{T}_0} x_{0,j} \langle \boldsymbol{a}_i, \boldsymbol{a}_j\rangle\right| \\ &\le \sum_{j\in\mathcal{T}_0} |x_{0,j}|\, |\langle \boldsymbol{a}_i, \boldsymbol{a}_j\rangle| \\ &\le \mu \sum_{j\in\mathcal{T}_0} |x_{0,j}| \\ &\le \mu\sqrt{S}\, \|\boldsymbol{x}\|_2 . \end{aligned}$$

Hence,

$$\max_{i\notin\mathcal{T}_0} |\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| \le \mu\sqrt{S}\, \|\boldsymbol{x}\|_2 . \quad [3]$$

ii For $i \in \mathcal{T}_0$,

$$\begin{aligned} |\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| &= \left|\sum_{j\in\mathcal{T}_0} x_{0,j} \langle \boldsymbol{a}_i, \boldsymbol{a}_j\rangle\right| \\ &\ge |x_{0,i} \langle \boldsymbol{a}_i, \boldsymbol{a}_i\rangle| - \left|\sum_{j\neq i} x_{0,j} \langle \boldsymbol{a}_i, \boldsymbol{a}_j\rangle\right| \\ &\ge |x_{0,i}| - \mu \sum_{j\neq i} |x_{0,j}| \\ &\overset{(a)}{\ge} |x_{0,i}| - \mu\sqrt{S}\, \|\boldsymbol{x}\|_2 , \end{aligned}$$

Note that $\max_{i\in\mathcal{T}_0} |x_{0,i}| \ge \frac{1}{\sqrt{S}} \|\boldsymbol{x}\|_2$. It holds that

$$\max_{i\in\mathcal{T}_0} |\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| \ge \tfrac{1}{\sqrt{S}} \|\boldsymbol{x}\|_2 - \mu\sqrt{S}\, \|\boldsymbol{x}\|_2 \quad [3]$$

iii Suppose that $\mu < \frac{1}{2S}$. Then

$\max_{i\in\mathcal{T}_0} |\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle| \ge \frac{1}{\sqrt{S}} \|\boldsymbol{x}\|_2 - \mu\sqrt{S}\, \|\boldsymbol{x}\|_2 > \mu\sqrt{S}\, \|\boldsymbol{x}\|_2 \ge \max_{i\notin\mathcal{T}_0} |\langle \boldsymbol{a}_i, \boldsymbol{y}\rangle|$.

One has $i_1 \in \mathcal{T}_0$. [3]

(b) The IHT algorithm is given by

Initialization:

$\boldsymbol{x}^0 = \mathbf{0}$.

Iteration: $\ell = 1, 2, \cdots$ until exit criterion true.

$$\boldsymbol{x}^\ell = H_S\left(\boldsymbol{x}^{\ell-1} + \boldsymbol{A}^T\left(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^{\ell-1}\right)\right).$$

[3]

i

$$\begin{aligned}\left\|\boldsymbol{r}^k\right\|_2 &\leq \frac{1}{2}\left\|\boldsymbol{r}^{k-1}\right\|_2 + 2.5\left\|\boldsymbol{w}\right\|_2 \\ &\leq \frac{1}{4}\left\|\boldsymbol{r}^{k-2}\right\|_2 + 2.5\left(1+\frac{1}{2}\right)\left\|\boldsymbol{w}\right\|_2 \\ &\leq \cdots < \frac{1}{2^k}\left\|\boldsymbol{r}^0\right\|_2 + 5\left\|\boldsymbol{w}\right\|_2 .\end{aligned}$$

Therefore, $c_k = 2^{-k}$ and $c' = 5$. [3]

ii If $k > \left\lceil \log_2 \frac{\|\boldsymbol{x}_0\|_2}{10^{-6}} \right\rceil$, then $\left\|\boldsymbol{r}^k\right\|_2 \leq 10^{-6}$. [2]